

i. (a) Given: $B = \sum_{i=1}^n m_i y_i$

$$m_i = \frac{x_i - \bar{x}}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$E(B) = E\left(\sum_{i=1}^n m_i y_i\right) = \sum_{i=1}^n m_i E(y_i)$$

Substitute: $E(y_i) = \alpha + \beta x_i$:

$$E(B) = \sum_{i=1}^n m_i (\alpha + \beta x_i) = \alpha \sum_{i=1}^n m_i + \beta \sum_{i=1}^n m_i x_i$$

$$E(B) = \alpha(0) + \beta(1) = \beta$$

Therefore, $E(B) = \beta$, this proves that B is an unbiased estimator of β .

(b) show that A can be written as a linear function of the y_i s. Then show that $E(A) = \alpha$.

$$A = \bar{y} - B\bar{x}$$

Since $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$ and $B = \sum_{i=1}^n m_i y_i$,

then $A = \frac{1}{n} \sum_{i=1}^n y_i - \bar{x} \sum_{i=1}^n m_i y_i$

Therefore, $A = \sum_{i=1}^n \left(\frac{1}{n} - \bar{x} m_i \right) y_i$

$$E(A) = E(\bar{y} - B\bar{x}) = E(\bar{y}) - \bar{x} E(B)$$

$$E(B) = \beta$$

$$E(\bar{y}) = E\left(\frac{1}{n} \sum_{i=1}^n y_i\right)$$

$$= \frac{1}{n} \sum_{i=1}^n E(y_i)$$

$$= \frac{1}{n} \sum_{i=1}^n (\alpha + \beta x_i)$$

$$= \frac{1}{n} \left(n\alpha + \beta \sum_{i=1}^n x_i \right)$$

$$\text{Since: } \sum_{i=1}^n x_i = n\bar{x}$$

$$\text{then } E(\bar{y}) = \alpha + \beta \bar{x}$$

$$\text{Therefore, } E(A) = (\alpha + \beta \bar{x}) - \bar{x}(\beta)$$
$$= \alpha + \beta \bar{x} - \beta \bar{x}$$

$$= \alpha$$

$$E(A) = \alpha$$

Therefore, A is unbiased estimator of α .

$$\therefore E(A) = \alpha, \quad E(B) = \beta$$

2. [Hint: $V(B) = \sum_{i=1}^n m_i^2 V(y_i)$]

Slope estimator B

$$B = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \sum_{i=1}^n m_i y_i$$

where $m_i = \frac{x_i - \bar{x}}{\sum_{j=1}^n (x_j - \bar{x})^2}$

Derivation of $V(B)$:

variance operator to $B = \sum_{i=1}^n m_i y_i$:

$$V(B) = V\left(\sum_{i=1}^n m_i y_i\right)$$

$$V(B) = \sum_{i=1}^n m_i^2 V(y_i)$$

Constant variance ($V(y_i) = \sigma^2$):

$$V(B) = \sigma^2 \sum_{i=1}^n m_i^2$$

Substitute $\sum_{i=1}^n m_i^2 = \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2}$

$$V(B) = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Intercept Estimator (A)

$$A = \bar{y} - B\bar{x}$$

A as a linear combination of y_i :

$$A = \frac{1}{n} \sum_{i=1}^n y_i - \bar{x} \sum_{i=1}^n m_i y_i = \sum_{i=1}^n \left(\frac{1}{n} - \bar{x} m_i \right) y_i$$

$$\text{Let } k_i = \frac{1}{n} - \bar{x} m_i.$$

Derivation of $V(A)$:

$$V(A) = V\left(\sum_{i=1}^n k_i y_i\right)$$

by Independence:

$$V(A) = \sum_{i=1}^n k_i^2 V(y_i)$$

by constant variance ($V(y_i) = \sigma^2$):

$$V(A) = \sigma^2 \sum_{i=1}^n k_i^2$$

Expand $\sum_{i=1}^n k_i^2$:

$$\sum_{i=1}^n k_i^2 = \sum_{i=1}^n \left(\frac{1}{n} - \bar{x} m_i \right)^2 = \sum_{i=1}^n \left(\frac{1}{n^2} - \frac{2\bar{x} m_i}{n} + \bar{x}^2 m_i^2 \right)$$

$$\sum_{i=1}^n k_i^2 = \frac{n}{n^2} - \frac{2\bar{x}}{n} \sum_{i=1}^n m_i + \bar{x}^2 \sum_{i=1}^n m_i^2$$

Since $\sum m_i = 0$ and $\sum m_i^2 = \frac{1}{\sum (x_i - \bar{x})^2}$

$$\sum_{i=1}^n k_i^2 = \frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Substitute into $V(A)$:

Answer: $V(A) = \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right]$

3. Linear transformation of X and Y in simple regression.

(a) Given:

$$X' = 10(X - 1)$$

Thus, $B' = \frac{B}{10}$

$$SE(B') = \frac{SE(B)}{10}$$

$$t'_0 = \frac{B'}{SE(B')}$$

$$t'_0 = \frac{B/10}{SE(B)/10}$$

$$t'_0 = \frac{B}{SE(B)} = t_0$$

The t -value stays the same

(b) Given:

$$Y'' = 5(Y + 2)$$

$$\text{Therefore, } B'' = 5B$$

$$SE(B'') = 5SE(B)$$

$$t''_0 = \frac{B''}{SE(B'')}$$

$$t''_0 = \frac{5B}{5SE(B)}$$

$$t''_0 = \frac{B}{SE(B)} = t_0$$

The t -value remains the same also.

(c) The linear transformations change units of the regression coefficient and its standard error. Both the point estimate (B) and its standard error ($SE(B)$), so:

$$t_0 = \frac{B}{SE(B)} \text{ remains unchanged.}$$

Therefore, hypothesis tests for the slope are completely unaffected by linear transformations.

The endpoints of the confidence interval B scale directly by the transformation factor $\frac{c}{a}$.

The coverage probability $(1 - \alpha)$ and statistical inference remain identical.